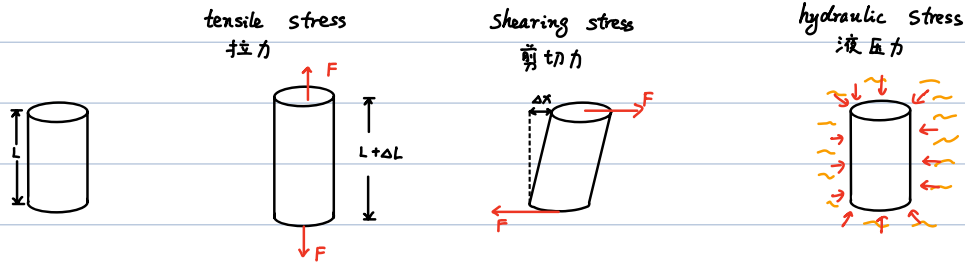


Elasticity

Solids are NOT perfectly rigid:



Stress: deforming force per area

Strain: unit deformation

modulus of elasticity:

$$\text{Stress} = \text{modulus} \times \text{strain} \quad (\text{for small stress})$$

1°. Tension & Compression

$$\text{stress: } F/A$$

$$\text{strain: } \Delta L/L$$

$$\text{Young's modulus: } \frac{F}{A} = E \frac{\Delta L}{L}$$

(杨氏模量)

2°. Shearing

$$\text{stress: } F/A$$

$$\text{strain: } \Delta x/L$$

$$\text{Shear modulus: } \frac{F}{A} = G \frac{\Delta x}{L}$$

(剪切模量)

3°. Hydraulic Stress

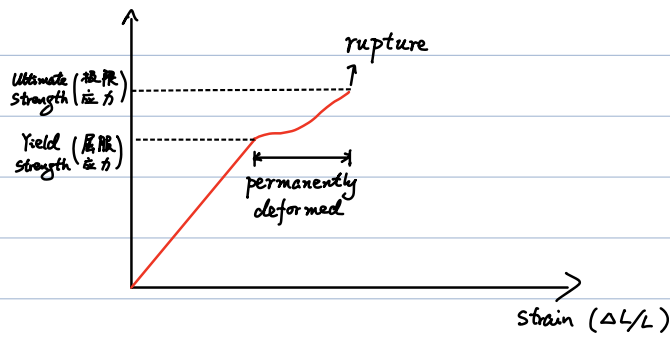
$$\text{Hydraulic stress: pressure}$$

$$\text{strain: } \frac{\Delta V}{V}$$

$$\text{Bulk modulus: } P = B \frac{\Delta V}{V}$$

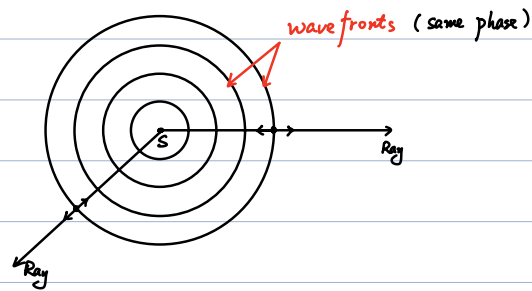
(体积模量)

Stress - strain curve



Sound Waves

Definition: any (mechanical) longitudinal waves



Speed of sound

previously, phonon in solids:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = v^2 \frac{\partial^2 u}{\partial x^2} \\ v = a \sqrt{\frac{K}{m}} \end{cases}$$

Sounds in air: Compression/expansion of air

$$v = \sqrt{\frac{B}{\rho}}$$

$$\begin{cases} \rho_{\text{water}} \approx 10^3 \text{ kg/m}^3 \\ \rho_{\text{air}} \approx 1.3 \text{ kg/m}^3 \end{cases}$$

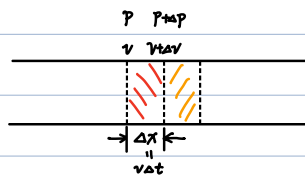
Naively, would get $v_{\text{air}} \gg v_{\text{water}}$ X

$$\begin{cases} B_{\text{water}} \approx 2.1 \times 10^9 \text{ Pa} \\ B_{\text{air}} \approx 1.4 \times 10^5 \text{ Pa} \end{cases} \quad (\text{more incompressible})$$

$$\Rightarrow v_{\text{water}} \gg v_{\text{air}}$$

$$\sim 1450 \text{ m/s} \quad \sim 330 \text{ m/s}$$

Derivation for $v = \sqrt{\frac{B}{\rho}}$:



$$(\Delta V < 0)$$

$$F = ma$$

$$\Rightarrow -\Delta p A = (\rho \Delta x A) \frac{\Delta v}{\Delta t}$$

$$\frac{\Delta x}{\Delta t} = v$$

$$\Rightarrow -\Delta p = \rho v \Delta v$$

air of volume $V = \Delta x t$ being compressed:

$$\frac{\Delta V}{V} = \frac{A \Delta v \Delta t}{A v \Delta t} = \frac{\Delta v}{v}$$

$$\Rightarrow -\Delta p = \rho v^2 \frac{\Delta v}{v}$$

$$\Rightarrow \rho v^2 = -\frac{\Delta p}{\Delta V/V} = B$$

$$\Rightarrow v = \sqrt{\frac{B}{\rho}}$$

Intensity :

$$I = \frac{P}{A}$$

P: energy transfer rate

A: area

For a sinusoidal sound wave :

$$S(x, t) = S_m \cos(kx - \omega t) \quad S: \text{longitudinal displacement}$$

we have

$$I = \frac{1}{2} \rho v \omega^2 S_m^2$$

Proof:

For a slice of air w/ thickness dx , area A

$$dK = \frac{1}{2} \rho dx A v_s^2$$

$$v_s = \frac{\partial S(x, t)}{\partial t} = \omega S_m \sin(kx - \omega t)$$

$$\Rightarrow dK = \frac{1}{2} \rho dx A \cdot \omega^2 S_m^2 \sin^2(kx - \omega t)$$

$$\Rightarrow \frac{dK}{dt} = \frac{1}{2} \rho A v \omega^2 S_m^2 \sin^2(kx - \omega t)$$

$$\Rightarrow \left(\frac{dK}{dt} \right)_{avg} = \frac{1}{4} \rho A v \omega^2 S_m^2$$

③

Without proof: $\left(\frac{dU}{dt}\right)_{\text{avg}} = \left(\frac{dK}{dt}\right)_{\text{avg}}$

$$\Rightarrow I = \frac{d(K+U)/dt}{A} = \frac{1}{2} \rho v \omega^2 s_m^2$$

For a point source that emits sounds isotropically,

$$I = \frac{P_s}{4\pi r^2} \quad (\text{assuming energy conservation})$$

Sound level: (声级, 噪声级)

$$\beta = 10 \lg\left(\frac{I}{I_0}\right) \quad \text{dB (decibel)}$$

where

$$I_0 \equiv 10^{-12} \text{ W/m}^2 \quad (\text{hearing threshold})$$

$$(I = I_0 \Rightarrow \beta = 0)$$

Note: $\beta_2 - \beta_1 = 10 \Leftrightarrow I_2/I_1 = 10$!

Bedroom at night $\sim 30 \text{ dB}$

Conversation $\sim 60 \text{ dB}$

Rock concert $\sim 110 \text{ dB}$

Pain threshold $\sim 120 \text{ dB}$

Note:

Sound level $\neq$ Sound frequency

$$\left\{ \begin{array}{l} \text{infrasonic (次声)} : < 20 \text{ Hz} \\ \text{audible frequencies} : 20 \text{ Hz} \sim 20 \text{ K Hz} \\ \text{ultrasound} (> 20 \text{ K Hz}) \end{array} \right.$$

Doppler effect

$$f' = f \frac{v \pm v_D}{v \pm v_S}$$

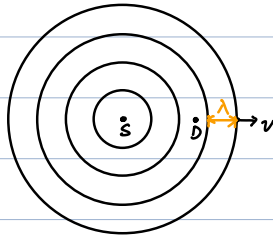
v : speed of sound in medium

v_D : speed of detector (relative to medium)

v_S : speed of source (relative to medium)

$\left\{ \begin{array}{l} \text{D \& S moves towards each other: } f' > f \\ \text{--- --- --- away from ---: } f' < f \end{array} \right.$

1. both stationary



f : # of wavefronts passing D in unit time:

$$f = \frac{vt/\lambda}{t} = \frac{v}{\lambda}$$

Reminder:

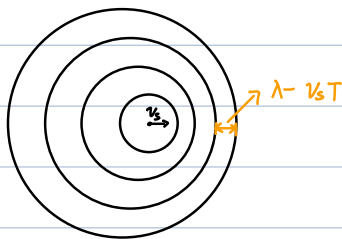
$$v = \frac{w}{k} = \frac{2\pi f}{2\pi/\lambda}$$

2. Source stationary, Detector moving

$$f' = \frac{(v \pm v_0)t/\lambda}{t} = \frac{v \pm v_0}{\lambda} = f \frac{v \pm v_0}{v}$$

3. Detector stationary, source moving

(towards D)



$$\begin{aligned} f' &= \frac{vt/\lambda'}{t} \\ &= \frac{v}{vT - v_s T} \\ &= f \frac{v}{v - v_s} \end{aligned}$$

4. Both moving:

$$f' = f \frac{v \pm v_0}{v \pm v_s}$$

Example: Train passing a station, an observer at the station hear the siren

$$1000 \text{ Hz} \leq f \leq 1200 \text{ Hz}$$

Q: speed of train?

$$\begin{cases} f \frac{v}{v - v_s} = 1200 \text{ Hz} \\ f \frac{v}{v + v_s} = 1000 \text{ Hz} \end{cases}$$

$$v = 330 \text{ m/s}$$

$$\Rightarrow \frac{v + v_s}{v - v_s} = \frac{12}{10}$$

$$\Rightarrow v_s = \frac{v}{11} = 30 \text{ m/s}$$

Example. Cosmological redshift

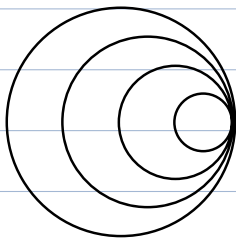
($\lambda \uparrow$, $f \downarrow$)

$\Rightarrow$ galaxies moving away from us

("big-bang" theory)

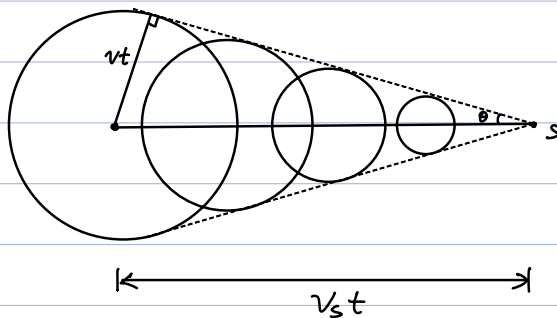
Supersonic speeds

$$v_s = v$$



$$\lambda - v_s T = 0$$

$v_s > v$ (Shock waves)



Envelop function form a cone:

Mach cone

Mach cone angle: $\theta = \arcsin \frac{v t}{v_s t} = \arcsin \frac{v}{v_s}$

Mach number: $\frac{v_s}{v}$

Examples of Shock waves: jet, bullet, boat, ...